ABDUL FOFANAH

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EDUCATION

University of York York, UK
BSc (Hons) Computer Science with Artificial Intelligence (with a Year in Industry) | On track for First-Class Honours Expected June 2028
Modules: Data Science (87%), Machine Learning & Optimisation (86%), OS, Security & Networking (83%), Software & Systems Engineering (83%)

Dagenham Park Church of England School Barking & Dagenham, UK | 2023
A-Levels: Economics (A), Maths (B), Computer Science (B), EPQ (A) | GCSEs: 10 grades 8–5, including Maths (7) & English Language (6)

EXPERIENCE

Industrial Placement Software Engineer June 2026 – Present
Leonardo UK Edinburgh, UK

- Designed and delivered a C++14 performance-testing capability for a distributed system, measuring latency, throughput and delivery reliability across TCP, UDP, synchronous communication, payload sizes and execution environments
- Resolved a native C++ Windows compatibility regression by replacing a mandatory load-time thread-diagnostics API dependency with runtime symbol resolution and graceful fallback, restoring automated test execution across supported Windows environments
- Isolated a transport-layer regression across multiple repositories using controlled A/B testing, Git submodule analysis and git bisect, narrowing the fault to a specific source change and validating the root cause through targeted rollbacks and CI
- Redesigned UDP performance-event correlation and recorder lifecycle using publisher-assigned sequence identifiers, C++ serialisation and callback-driven start/stop coordination, preventing mispaired events and incomplete measurements under out-of-order delivery
- Integrated and validated automated performance tests in Jenkins CI, using isolated runs and log analysis to diagnose timeout failures and adapt workloads to fixed pipeline limits, ensuring associated test definitions passed the integration suite
- Built Grafana dashboards using SQL queries to visualise p95 latency, throughput, batch duration, payload scaling and transport comparisons, enabling repeatable analysis of performance-test results
- Reworked Markdown onboarding documentation for WSL2, Podman and VS Code developer-environment setup, standardising installation and configuration guidance used to onboard a new engineer

Project Engineering Manager – Industrial Placement Team August 2026 – Present
Leonardo UK Edinburgh, UK

- Led the team's delivery of system requirements and functional models across systems, software and hardware, allocating tasks, managing dependencies and contributing directly to the work, with the requirements successfully passing System Requirements Review
- Authored and maintained the Project Engineering Plan, defining scope, responsibilities, lifecycle activities, integration, verification and risk, and incorporating review feedback throughout the project
- Translated stakeholder needs and operational scenarios into use cases, functional models and system-level requirements using IBM DOORS, Confluence and draw.io, maintaining traceability through review
- Developed functional models and flow dictionaries using verb–noun decomposition as well as information, control and energy flows, supporting requirements development, interface identification and the emerging system architecture
- Planned how system requirements would be verified during integration, selecting appropriate verification methods and identifying the evidence and test equipment needed to demonstrate compliance

Earlier Experience – Technology & Web Development 2022 | Remote

- ***Cisco*** | Co-developed and pitched a winning, research-backed IoT health solution to Cisco professionals in a competitive innovation challenge
- ***Kainos*** | Developed a responsive HTML/CSS interface and completed OWASP Juice Shop labs on SQL injection and cross-site scripting (XSS)
- ***Ezoic*** | Built and maintained a live WordPress site, customising templates and plugins for performance, accessibility and SEO

PROJECTS

Oxford-IIIT Pet Image Classifier – ML & Optimisation Coursework (92%) | Python · PyTorch · Torchvision · NumPy · scikit-learn | GitHub

- Designed and trained a 4.37M-parameter VGG-style Convolutional Neural Network in PyTorch for 37-class Oxford-IIIT Pet classification, using convolution, batch normalisation, pooling, dropout and adaptive pooling
- Engineered a four-channel RGB + trimap-mask pipeline with synchronised augmentation, stratified train/validation splitting and dataset-specific normalisation across 3,496 training images
- Optimised training with AdamW, label-smoothed cross-entropy and OneCycleLR over 29 epochs, achieving 73.62% accuracy on the test set

Market Data Pipeline – Personal Project | C++17 · Java · Python · Gson · Pandas · NumPy · Matplotlib | GitHub

- Built an end-to-end market-data pipeline across C++17, Java and Python, ingesting OHLCV CSV data into JSONL, processing it into trading-signal data and producing analytical visualisations
- Implemented configurable rolling-window analysis in Java, computing moving averages, standard deviation and z-scores to generate trading signals, with Python/Pandas and Matplotlib used for summary statistics and time-series visualisation

Java Game Development – Software Engineering Coursework (83%) | Java · LibGDX · UML · GitHub Actions · Gradle · CI/CD | GitHub

- Developed and extended a LibGDX Java game across greenfield and brownfield development, using OOP and UML to meet a product brief
- Configured GitHub Actions CI to automate Gradle builds, test execution, Javadocs and runnable JAR packaging

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Bash, HTML/CSS, Markdown
ML & Data: PyTorch, Torchvision, scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Jupyter Notebook
Systems & DevOps: Linux, WSL2, Windows API, TCP/IP, UDP, CMake, Ninja, Clang, Podman, Jenkins, GitHub Actions, CI/CD
Tools & Platforms: Git/GitHub, Bitbucket, Jira, Confluence, IBM DOORS, draw.io, Grafana, Visual Studio Code

LEADERSHIP & ACTIVITIES

- **University of York Boxing Club – Boxer & Merchandise Secretary** | Built a 2–0 amateur record after taking up boxing at university
- **Student Outreach** | Returned to former sixth form to speak to students about university, degree apprenticeships and workplace pathways
- **Astra Collective – Charity Runway Model** | Modelled in two university runway shows, including a fundraiser for the Pink Ribbon Foundation